

1. Administrative & Study Identification

Full Study Title: A Study of the Epidemiology and Hospital Management of Patients With PROS in France: a PMSI Study **Short Title/Acronym:** EpiPROS **Protocol Number:** CBYL719F1FR02 **NCT Number:** NCT07222423 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** N/A (Non-interventional/Observational Database Study) **Phase of Development:** N/A (Observational) **Medical Monitor:** Novartis Medical Affairs Lead, France

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the epidemiology, hospital management, and associated healthcare costs of patients diagnosed with PIK3CA-Related Overgrowth Spectrum (PROS) in France using the National Hospital Discharge Database (PMSI). **Secondary Objectives:** To characterize the clinical manifestations, patient journey, treatment patterns, and Healthcare Resource Utilization (HCRU) of the PROS patient population within the French hospital setting.

2.2 Background & Rationale To address the gap in real-world data regarding the incidence, prevalence, and clinical burden of PIK3CA-Related Overgrowth Spectrum (PROS), a rare genetic disorder. By utilizing the French national health data system (SNDS/PMSI), this study aims to comprehensively describe the hospital management pathways and economic impact of PROS to better inform clinical guidelines and healthcare policies.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Retrospective Cohort / Database Study) **Control Type:** Single-arm (Descriptive cohort) **Blinding:** N/A (Open-label/Database analysis) **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** Dependent on database query (Population P1 identified via comprehensive PMSI search algorithm) **Number of Sites:** N/A (Nationwide database study using French PMSI database) **Estimated Study Start:** 2025 **Estimated Primary Completion:** 2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients identified in the French National Hospital Discharge Database (PMSI). 2. Population P1: Patients identified using a combined query of ICD-10 diagnostic codes indicative of PROS-related conditions during the selection period. 3. Patients with at least one recorded hospital admission or outpatient hospital visit related to PROS management within the study timeframe.

4.2 Exclusion Criteria 1. Patients whose records lack sufficient longitudinal data to assess HCRU. 2. Patients who have formally exercised their right to opt-out of the secondary use of their personal health data in the SNDS/PMSI system. 3. Missing core demographic identifiers (e.g., age, sex) within the database.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Observational design assessing real-world management) **Route of Administration:** N/A **Duration of Treatment:** N/A (Retrospective longitudinal observation window)

5.2 Concomitant & Prohibited Therapy Permitted: All standard-of-care medical therapies and surgical interventions captured within the hospital database. **Prohibited:** N/A

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Index Date	Day 0 First recorded PMSI
diagnostic code for PROS	Baseline Period	-12 to -24 Months	
Retrospective assessment of patient history and comorbidities prior to index date	Follow-up Period	Variable	
(Longitudinal)	Assessment of hospitalizations, procedures, and related management pathways	End of Study	End of data availability
Final extraction of HCRU and cost metrics			

(Note: As a retrospective database study, there are no prospective patient visits or onsite activities)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Epidemiology metrics (incidence and prevalence estimates) of PROS in France, and description of hospital management components (e.g., frequency of surgical interventions, pharmacological treatments). **Measurement Method:** Analysis of PMSI longitudinal hospital discharge data and associated ICD-10 diagnostic/procedure codes.

7.2 Secondary & Exploratory Endpoints 1. Assessment of Healthcare Resource Utilization (HCRU), including length of hospital stay, intensive care admissions, and outpatient visits. 2. Direct medical costs associated with the hospital management of PROS patients from the perspective of the French healthcare system.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A (Routine pharmacovigilance AE reporting is not applicable due to the retrospective, de-identified database nature of the study). **Serious Adverse Events (SAEs):** N/A **Data Monitoring Committee (DMC):** N/A

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible patients identified in the PMSI database (Population P1). **Sample Size Justification:** Not powered for inferential hypothesis testing; maximum available sample size extracted from the national database to provide descriptive epidemiological precision. **Handling of Missing Data:** Descriptive statistics will be based on available data; missing data handling rules will be applied per standard real-world evidence (RWE) guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: The study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), French CNIL (Commission Nationale de l'Informatique et des Libertés) data privacy regulations, and Health Data Hub requirements (MR-006 methodology). **Monitoring Plan:** N/A (Data extraction and algorithmic validation by a designated contract research organization/data analyst team). **Audit Trail:** Strict data security, pseudonymization, and tracking of database extraction scripts compliant with European GDPR.

11. Study Identifiers & Governance

Primary ID: CBYL719F1FR02 **Registry Number:** NCT07222423 **Sponsor/Lead Organization:** Novartis Pharmaceuticals / Novartis France **Study Title:** EpiPROS **Official Regulatory Title:** Epidemiology and Hospital Management of Patients with PROS in France: a PMSI Study

12. Clinical Framework (Hyperkalemia Specific - Adapted for PROS)

Primary Condition: PIK3CA-Related Overgrowth Spectrum (PROS)
Diagnosis Threshold: Identification via specific ICD-10 codes / clinical diagnostic mapping
Medical Methodology: Multidisciplinary hospital management (Surgical and Medical standard of care)
Optimization Target: Characterization of real-world patient journey and clinical burden

13. Study Procedures & Methodology

Management Pathway: Real-world hospital management patterns tracking
Education Protocol (HCP): N/A
Education Protocol (Patient): N/A
Quality Audit Mechanism: Algorithmic validation of the PMSI data query (P1 population selection)

14. Observation & Follow-Up Schedule

Total Duration: Retrospective longitudinal observation (Extraction covers multi-year data through 2028)
Onsite Visit Cadence: N/A
Remote Monitoring Frequency: N/A
Checklist Review Interval: N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				